



Ahmedabad  
University

GROUP 17

# Probabilistic Adaptive Traffic Signal Control in Intelligent Transport Systems (ITS)

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# Project Context and Motivation

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

### Application Domain

- Intelligent Transportation Systems (ITS)

### System Recap

- 4-way, 8-directional traffic simulation
- Goal: Optimize signal schedules

### Why Uncertainty Matters

- Traffic is inherently random
- Vehicles arrive in unpredictable clumps
- Deterministic timers treat every minute as “average”
- Real-world  $\neq$  average world

# Deterministic Baseline Overview

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

### M3 Baselines

- Fixed-Timer
- Time-of-Day (ToD) Timer

### Performance

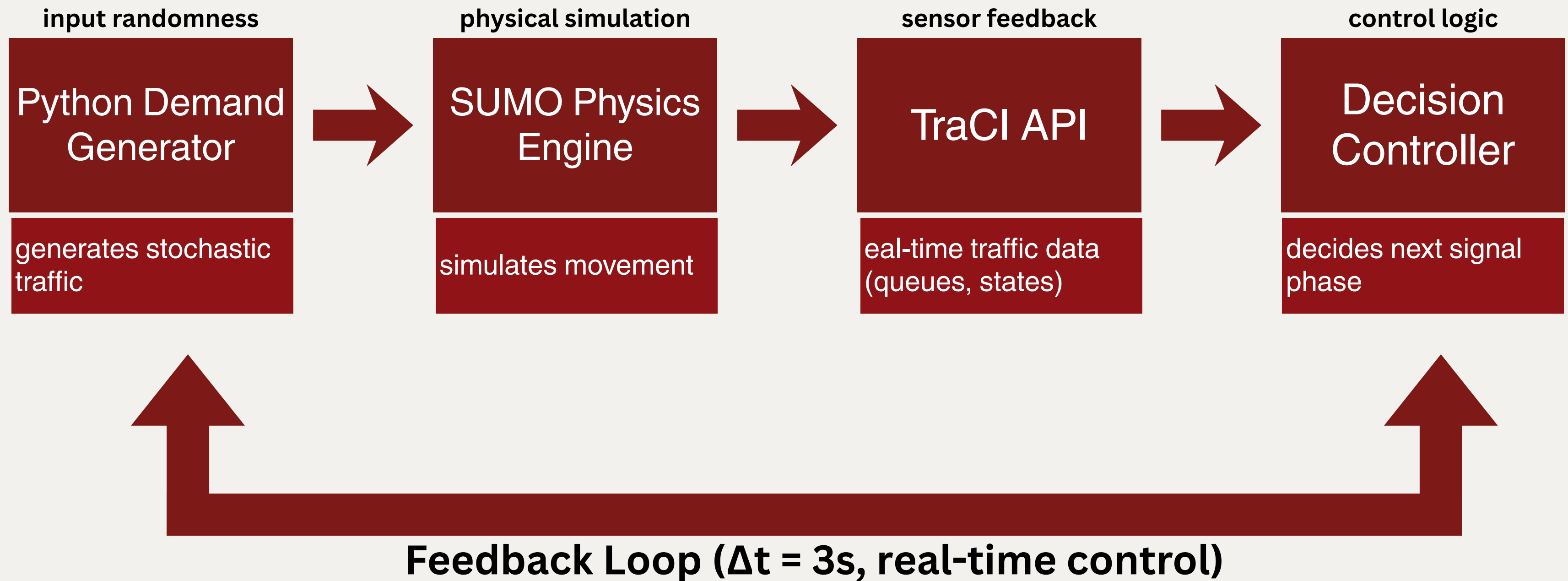
- Queue Reduction (ToD vs Fixed): -8.9%
- Variance Drop: 4.1%

### Limitations

- No real-time awareness
- Green time wasted on empty lanes
- Congestion builds during random surges

# System Pipeline Overview

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection



# Sources of Uncertainty

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

### SOURCES

#### Sensor Noise

- Detection limits
- Occlusion effects

#### Environmental Clumping

- Random traffic clusters
- Platoon formation

### IMPACT

#### Why Deterministic Models Fail?

- Cannot handle tail risks
- Fails during extreme surges
- Leads to permanent backlog





# Random Variables in the System

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection



### Arrivals ( $N$ )

- Discrete random variable
- Non-homogeneous Poisson arrivals



### Queue State ( $Q$ )

- Vehicles waiting per lane
- Observed via sensors



### Waiting Time ( $W$ )

- Total accumulated delay
- Across all vehicles

❏ **Key Assumption:** Vehicle arrivals are independent across lanes, though this assumption is challenged by real-world platooning effects.

# Randomized Algorithm Design

Probabilistic Phase Switching using weighted “virtual dice”

## Algorithmic Logic:

Decision Rule: Probability  $\propto$  Queue Length

1

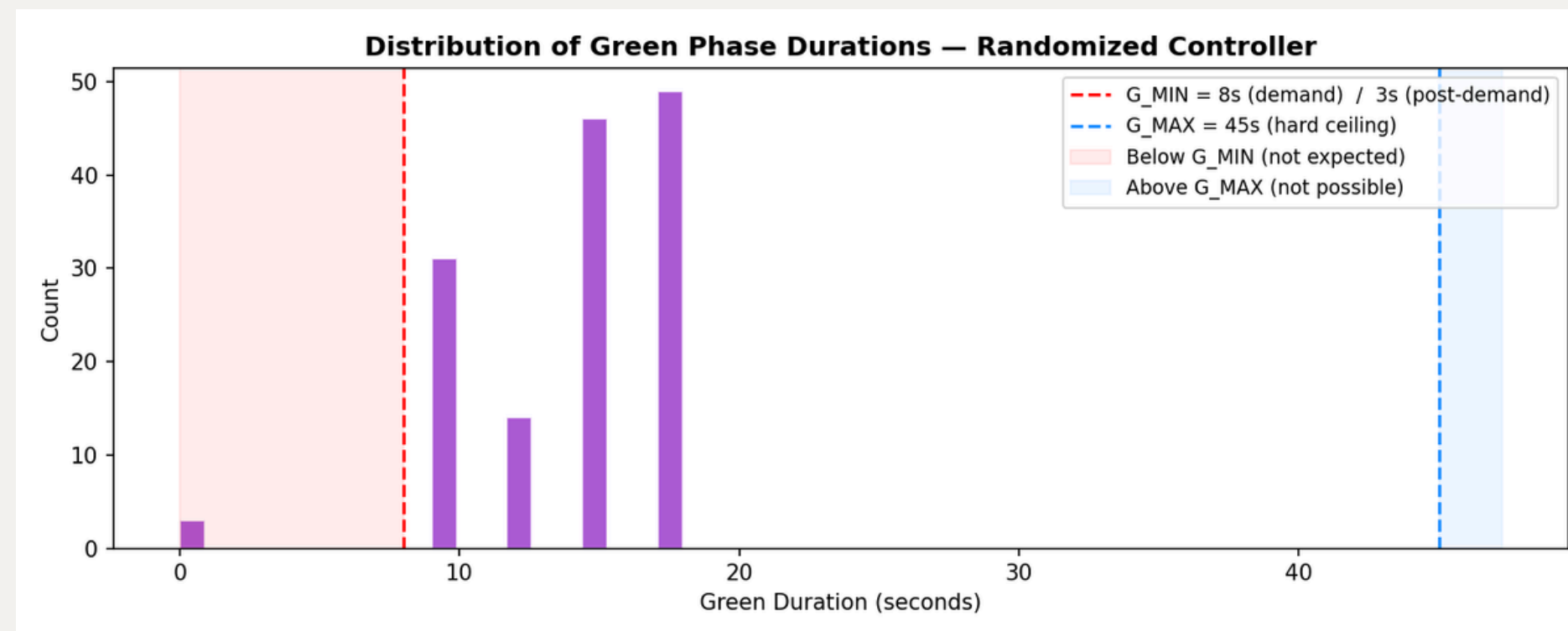
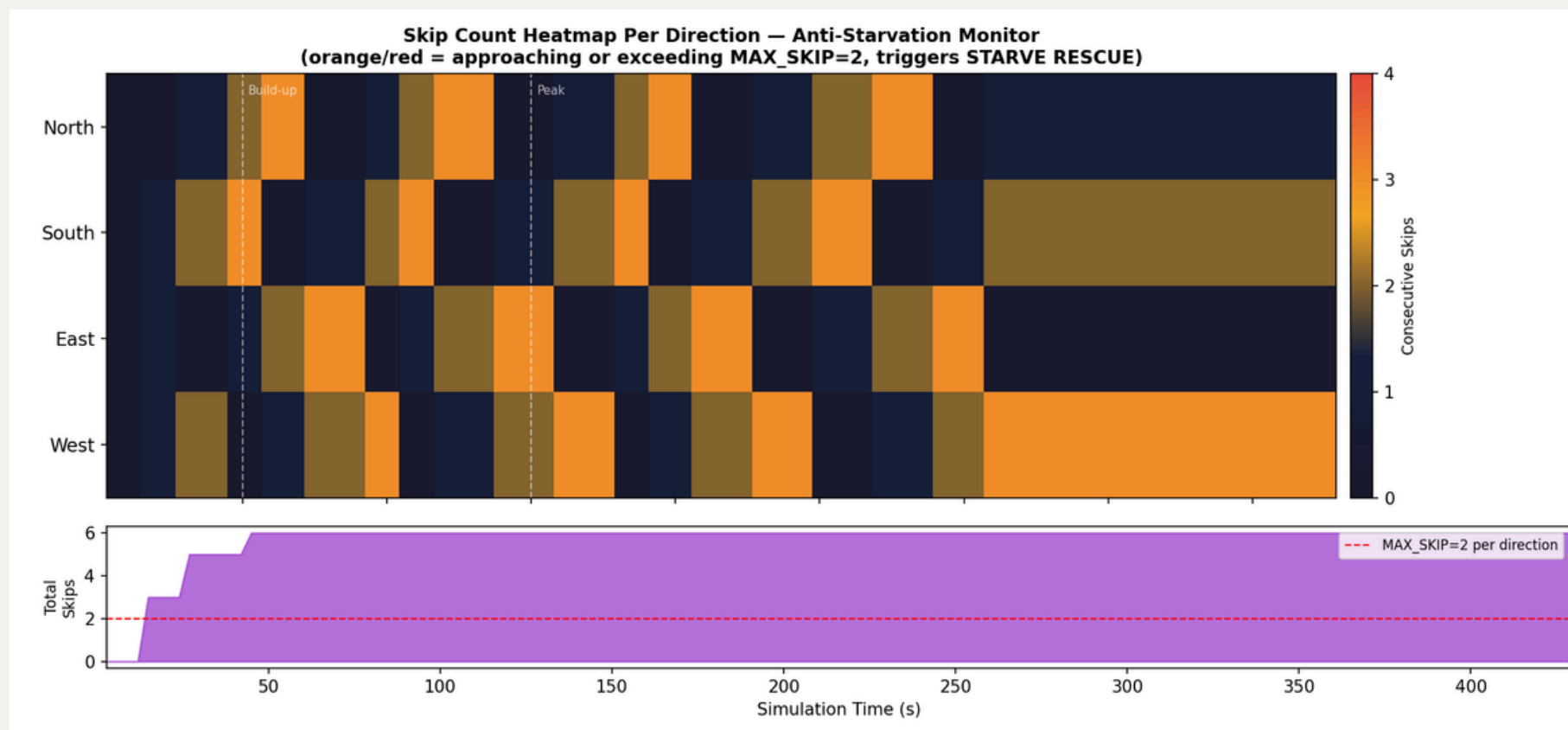
### Heavy Lanes

Higher probability of receiving Green phase

2

### Empty Lanes

Probabilistically skipped to optimize flow



# Probabilistic Reasoning Behind the Algorithm

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

### Breaks Traffic Resonance

- Breaks resonance in traffic cycles
- Adds controlled “jitter”

### Load Balancing

- Fair distribution of Green time
- No lane starvation

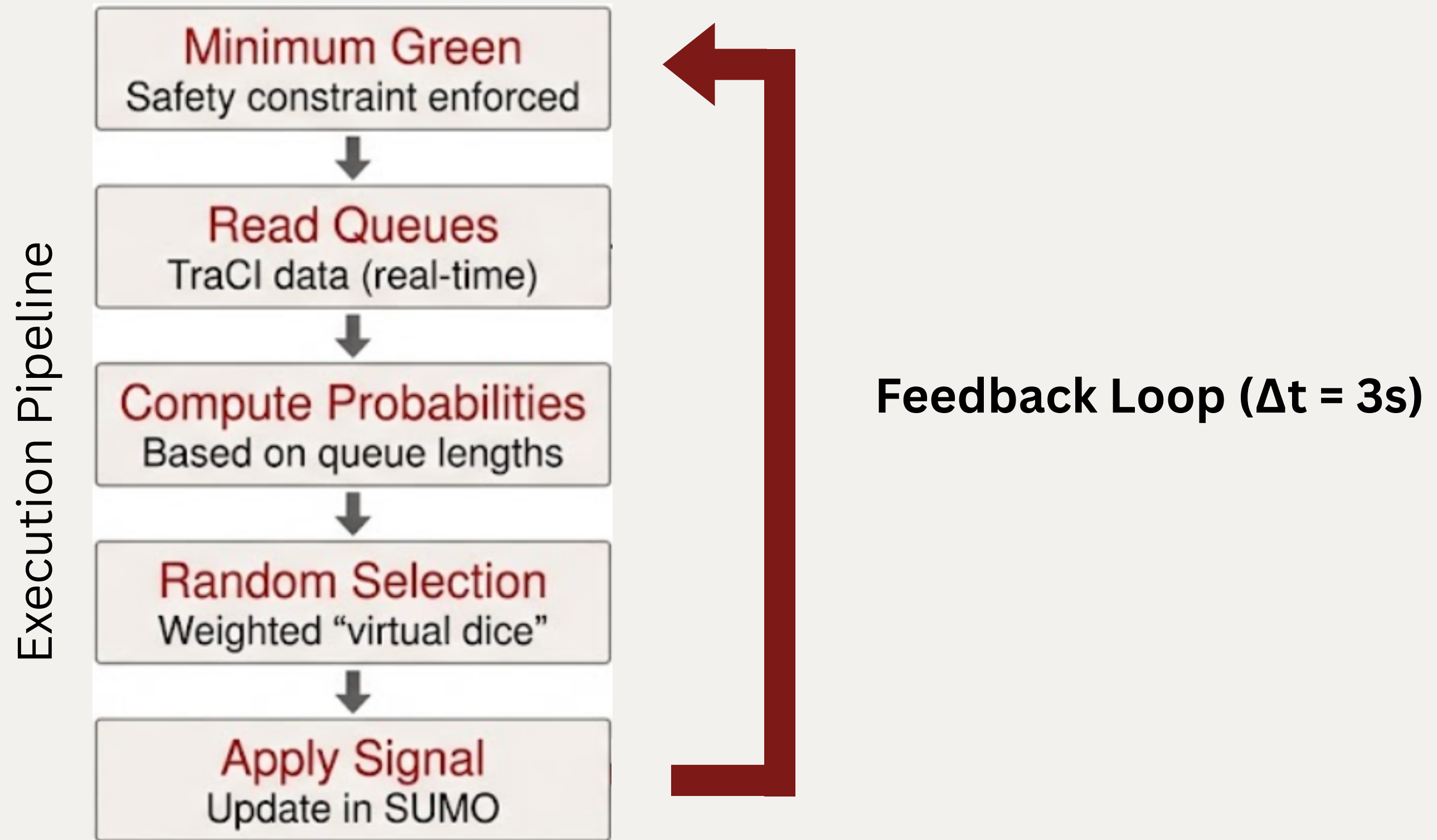
### Key Concept

- Small loss in predictability
- Large gain in stability



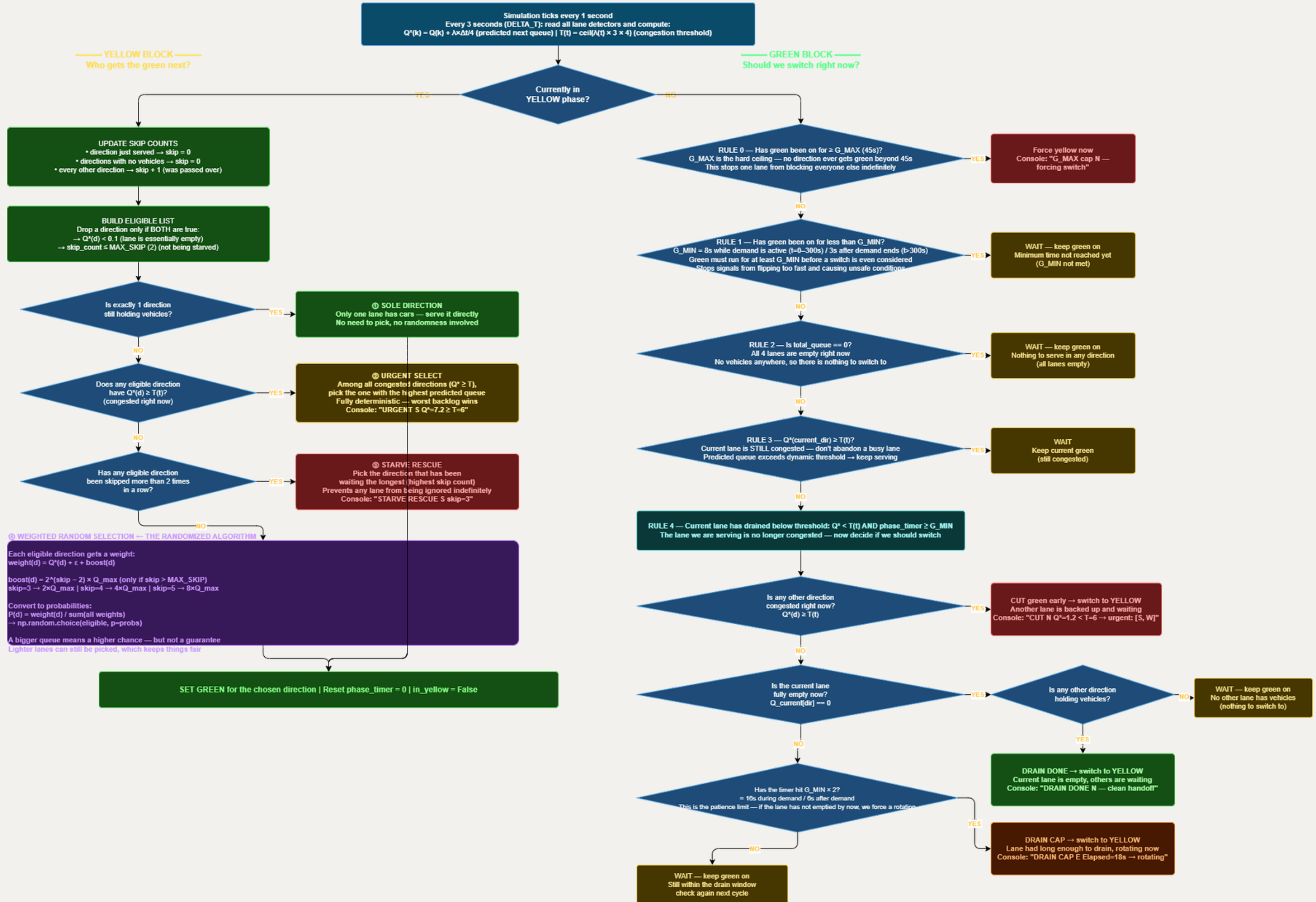
# Implementation of the Randomized Strategy

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection



# RANDOMIZED ADAPTIVE TRAFFIC SIGNAL — COMPLETE ALGORITHM FLOWCHART

Left column: YELLOW BLOCK (who gets green) | Right column: GREEN BLOCK (when to switch)



# Evaluation Metrics

## Tail Risk Analysis using Chernoff Bounds

**Mean and Peak Waiting Time:** Measures overall intersection efficiency and how well the algorithm handles worst-case extreme delays.

**Mean Queue and Variance:** Measures load balancing and intersection clear rates.

**Markov Transition Probabilities:** Measures the probability of escaping a congested state ( $P_{10}$ ) versus staying in a congested state ( $P_{11}$ ).

**Why These Are Appropriate:** They capture both the average driver's experience and the mathematical stability of the intersection.

# Evaluation Metrics

## Tail Risk Analysis using Chernoff Bounds

### ● Peak Hour Risk

$\lambda\Delta t = 1.50$  | High congestion risk

$P(N \geq 2.25) \leq 85.0\%$

$P(N \geq 3.00) \leq 56.0\%$

$P(N \geq 4.50) \leq 14.3\%$  (critical)

### Off-Peak

$\lambda\Delta t = 0.51$  | Low arrival rate

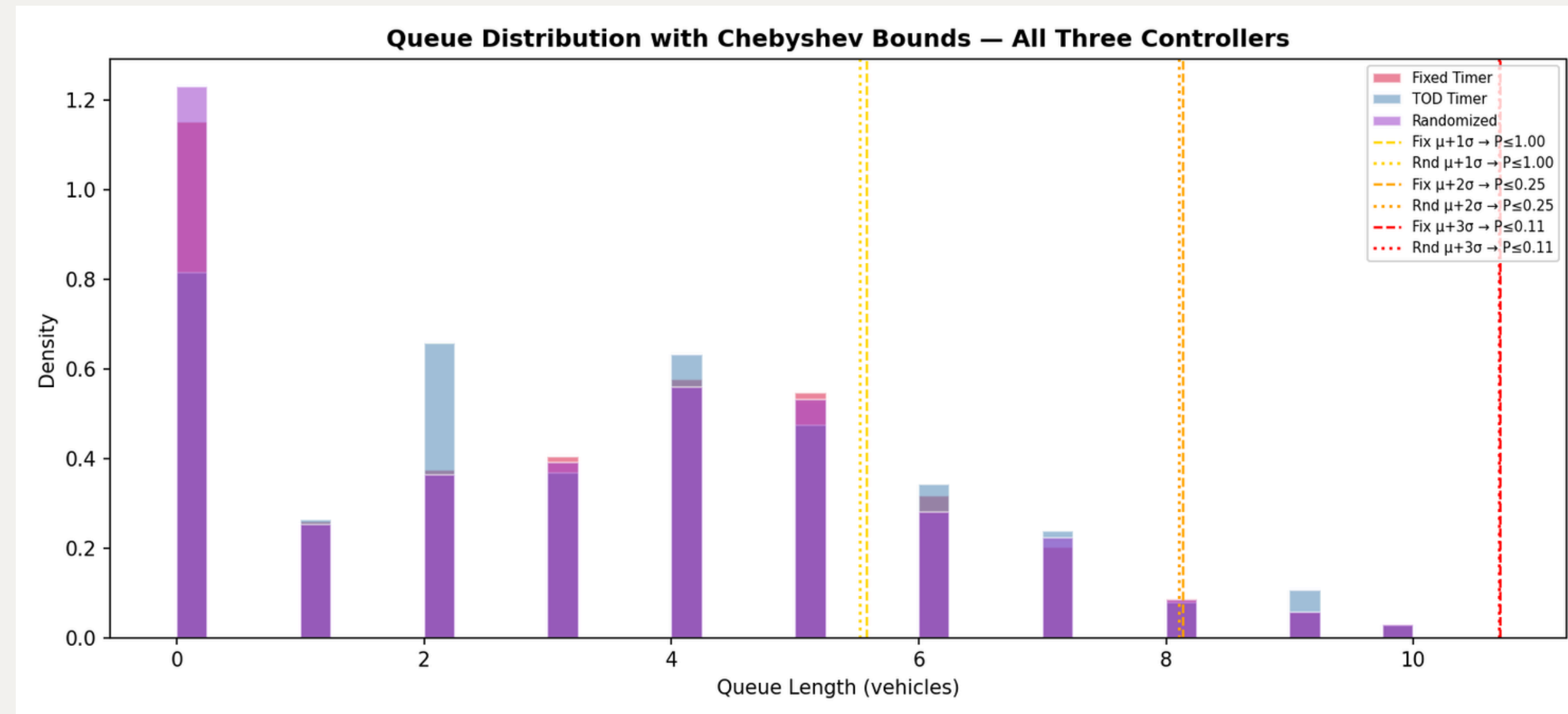
$P(N \geq 1.53) \leq 51.6\%$

### Build-Up

$\lambda\Delta t = 1.26$  | Moderate traffic

$P(N \geq 3.78) \leq 19.5\%$

**Chernoff Bound Analysis:** We calculated the probability of extreme surges (Tail Risks) that deterministic timers fail to handle.





# Comparative Results- Performance

## Mean Wait Time

Fixed: 447.4s | TOD: 648.2s | **Randomized: 281.1s**

## Peak Wait Time

Fixed: 1568s | TOD: 2896s | **Randomized: 1039s**

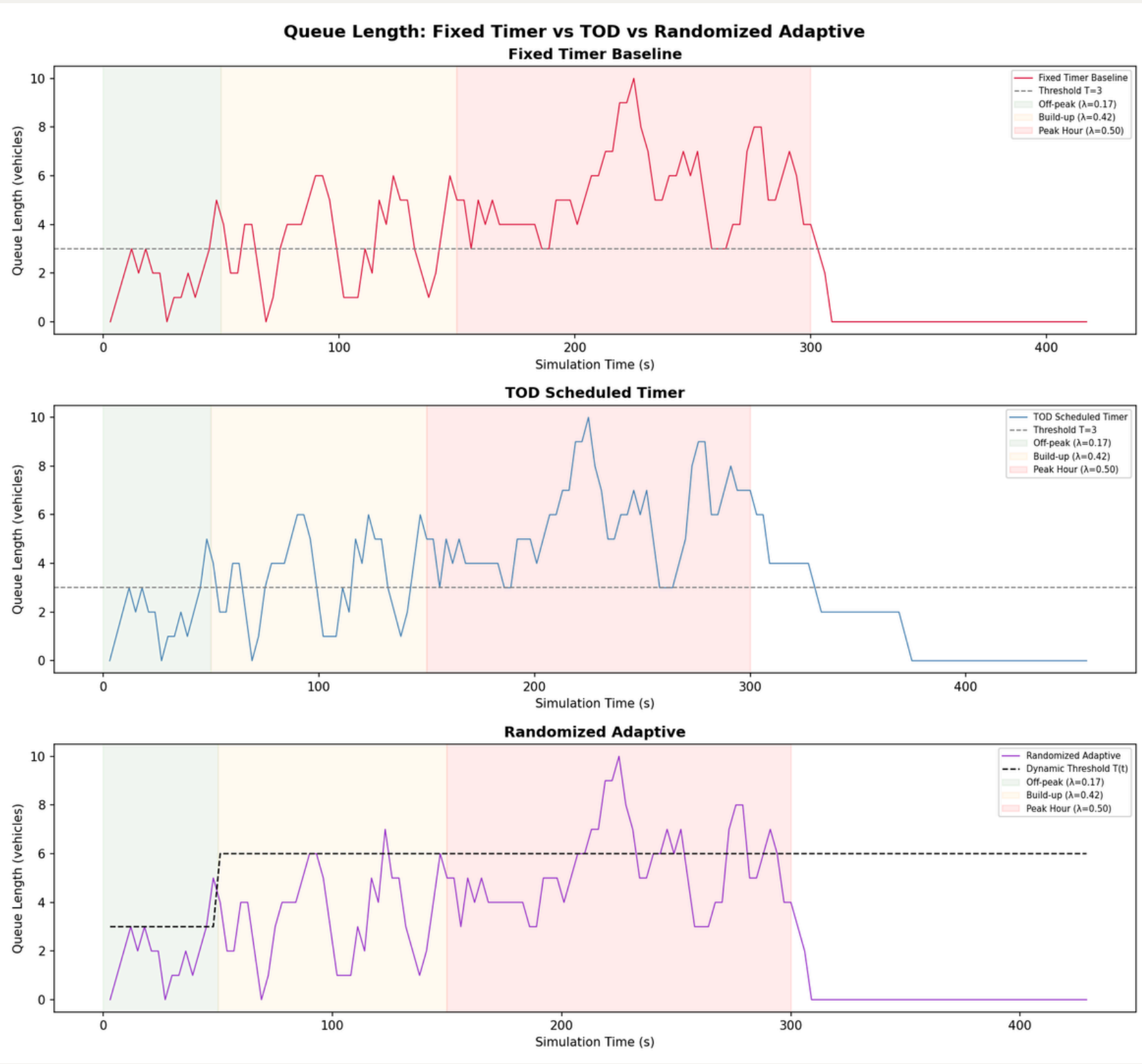
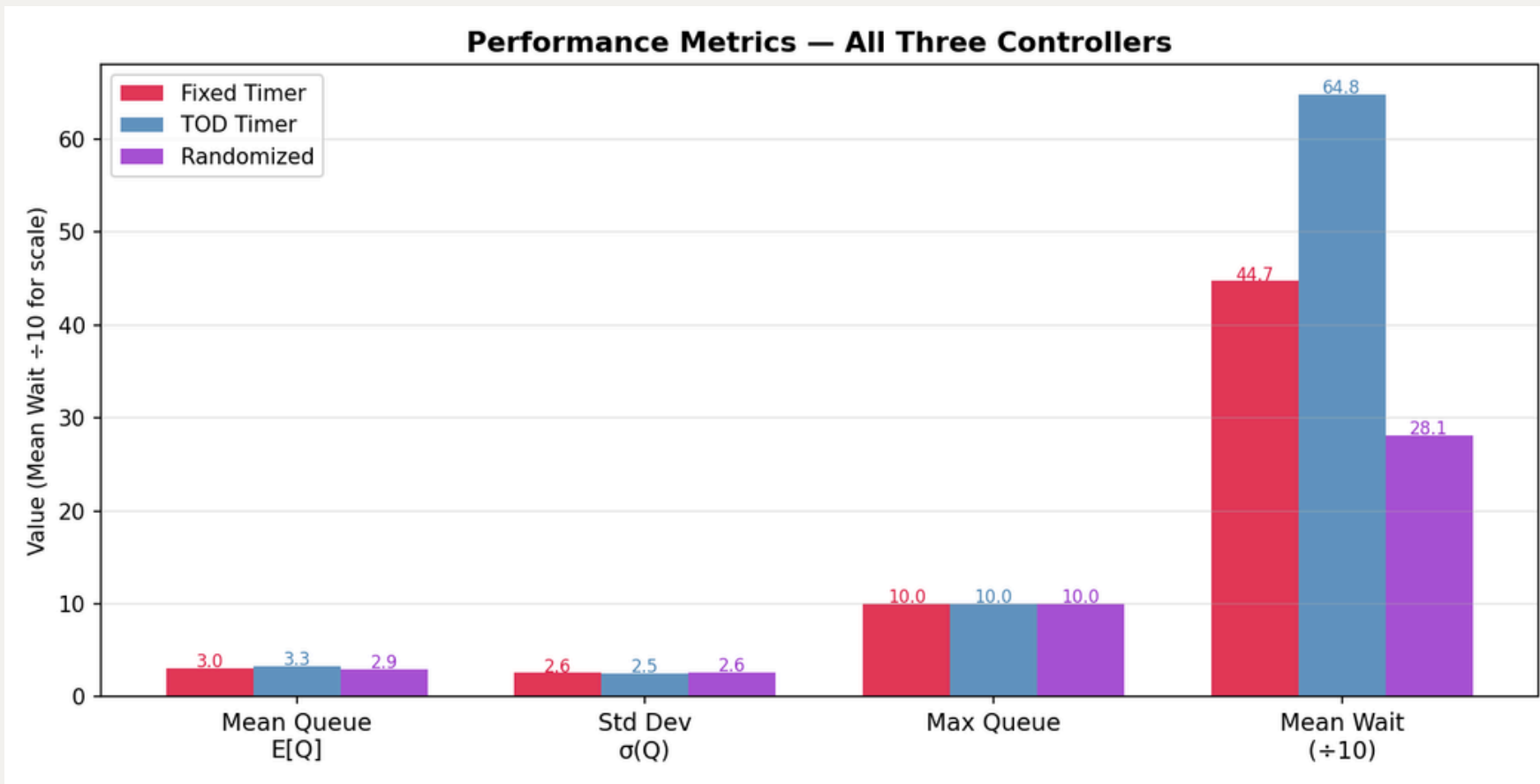
## Mean Queue

Fixed: 3.01 | TOD: 3.28 | **Randomized: 2.94**

## Variance

Fixed: 6.56 | TOD: 6.29 | **Randomized: 6.67**

**JENSEN GAP:** Fixed (6.56) | TOD (6.29) | **Randomized (6.67)**



# Comparative Results- Markov Transitions

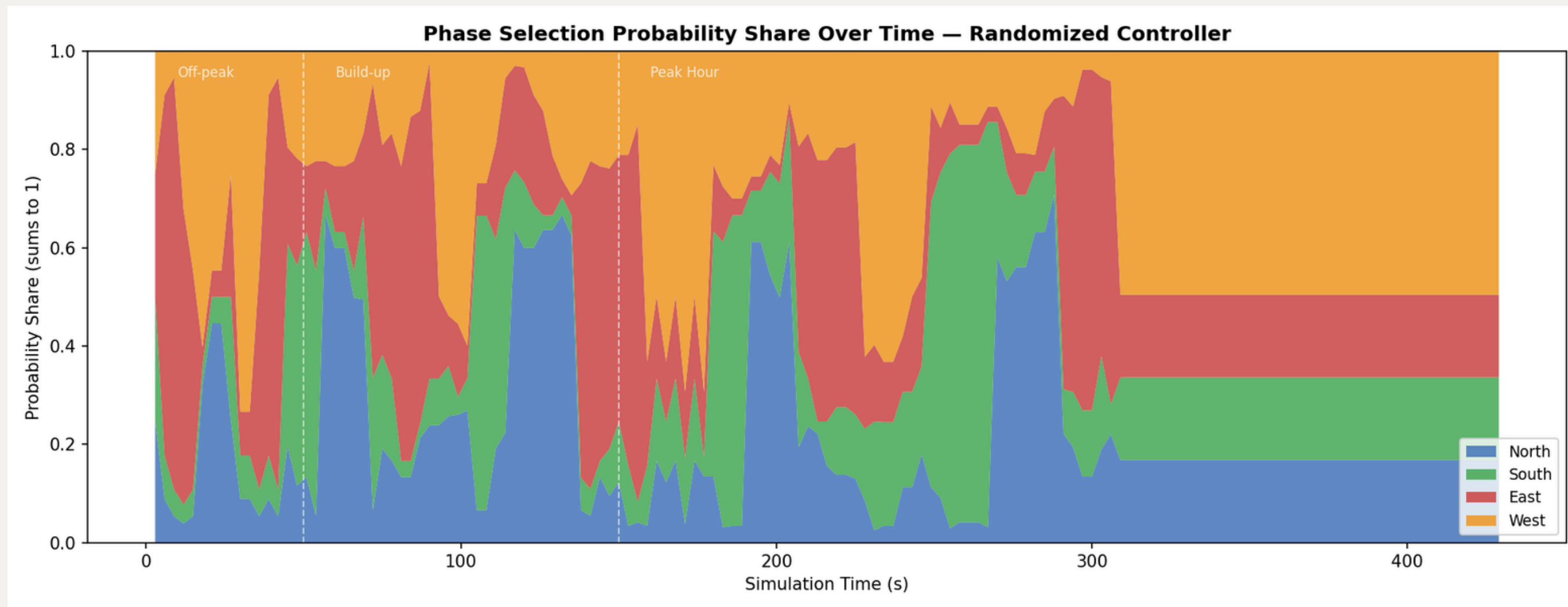
## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

**Fixed Timer Transitions:** P11 (Stays Jammed) = 0.896 | P10 (Clears Jam) = 0.104

**TOD Timer Transitions:** P11 (Stays Jammed) = 0.907 | P10 (Clears Jam) = 0.093

**Randomized Transitions:** P11 (Stays Jammed) = 0.643 | P10 (Clears Jam) = 0.357

**Observation:** The randomized system increases the chance of clearing a traffic jam by 3.4 times!



# Key Insights

## Stochastic Modeling of Traffic Flow at a Four-Way Intersection

### Main Observation 1:

The Randomized Adaptive strategy achieved a massive 37.2% reduction in Mean Wait Time compared to the baseline.

### Main Observation 2:

Time-of-Day (TOD) performed worse in peak conditions because adjusting a blind timer does not fix the underlying issue of traffic variance.

### Main Observation 3:

By dropping the P11 congestion probability to 0.643, the randomized system actively decongests the intersection.

### Final Conclusion

Controlled randomness is the cure for traffic resonance. By using a probability-weighted decision rule, we successfully mitigated extreme Poisson tail risks.

